Chapter 8 · Semester 1

STATICS

Notes by Cikgu Suhaili

15 worked examples 1 simulation Exercises with answers

8.1 Equilibrium of a particle

- **Statics** is the study of forces in equilibrium. In equilibrium all the forces on a body are balanced, so there is no net change in its motion.
- An object is in **equilibrium** when there is no net force on it.

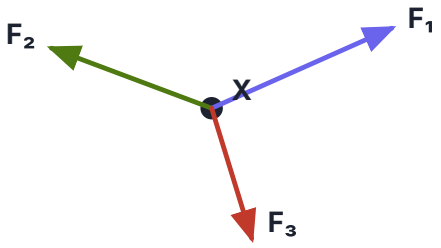
Type of equilibrium	Meaning
static equilibrium	at rest, $v = 0$
dynamic equilibrium	moving with constant velocity, $a = 0$
translational equilibrium	no linear acceleration
rotational equilibrium	no angular acceleration

Particle

is an object which has mass and is small enough to be regarded as a point.

Coplanar forces

are forces that lie in (act in) the same plane.



Concurrent forces

are forces whose lines of action intersect at a single point.

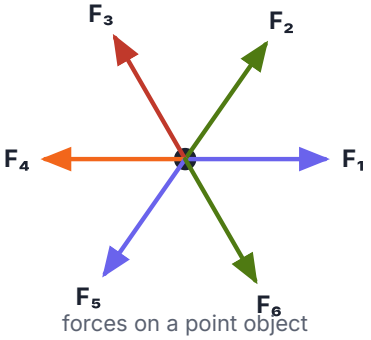
Condition for equilibrium of a particle

The **resultant (net) of the concurrent forces acting on the particle is zero**: $\Sigma\mathbf{F} = 0$, i.e. $\Sigma F_x = 0$ and $\Sigma F_y = 0$.

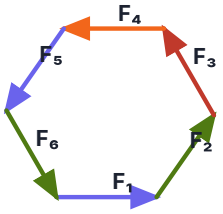
Two methods for a particle in equilibrium

- (i) **closed polygon of forces** (a scale drawing); (ii) **resolving forces into components**, using a free-body diagram (a diagram of a body showing all the forces acting on it).

> **Polygon of forces**
>
> A point object acted on by several forces is in equilibrium only if the force vectors, drawn head to tail, **form a closed polygon**.



forces on a point object



head to tail: a closed polygon

Draw arrows to show the magnitudes and directions of all the external forces.

Choose a scale, e.g. 1 cm : 2 N.

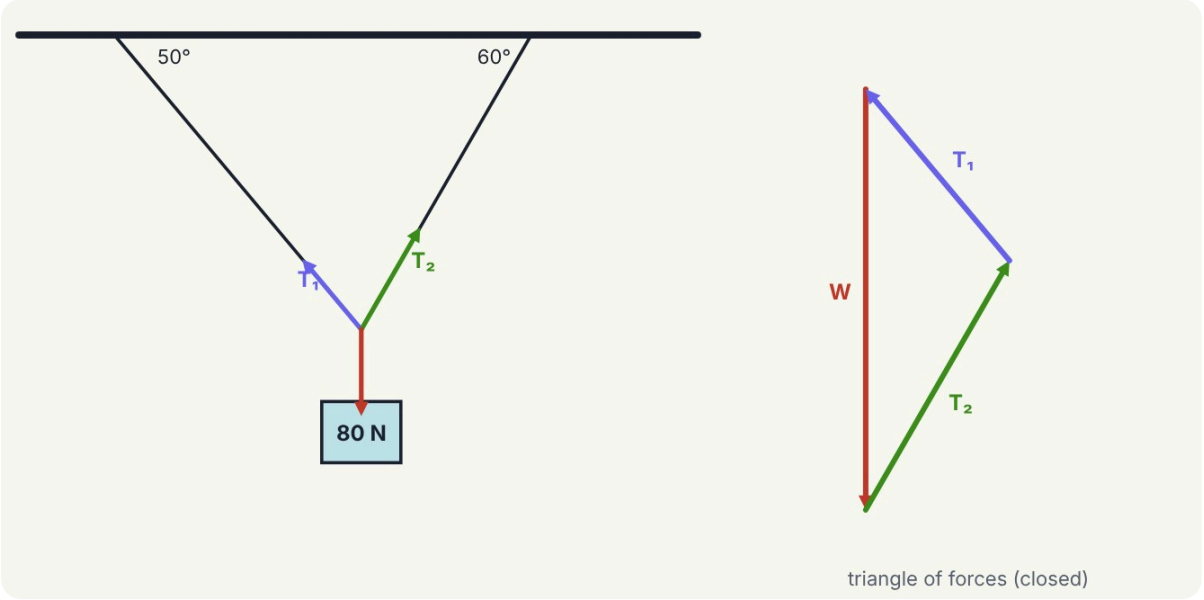
Add all the forces head to tail. The vector from the start to the end is the resultant (net) force; if the head of the last vector meets the tail of the first, the polygon is closed and the net force is zero.

Triangle of forces (STPM 8.1(e)): when a particle is in equilibrium under exactly three forces, the three force vectors drawn head to tail form a closed triangle.

Simulation

Three forces in equilibrium: the triangle of forces

A weight hangs from a knot held by two ropes. Change the angles and the weight: the tensions update, and the three forces drawn head to tail always close into a triangle.



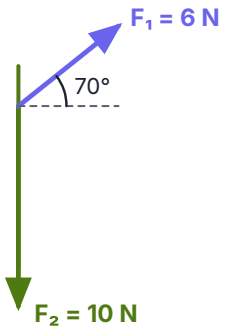
triangle of forces (closed)

T_1	T_2	$\Sigma Fx = T_2\cos - T_1\cos$	$\Sigma Fy = T_1\sin + T_2\sin - W$
42.6 N	**54.7 N**	**0.00 N**	**0.00 N**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 8 to use the simulation.

Example 8.1.1

A system of concurrent forces acts on a particle: F_1 = 6 N at 70° above the horizontal and F_2 = 10 N vertically downward. Find the force F_3 that keeps the particle in equilibrium.



Solution

Closed polygon (1 cm : 2 N): the closing side measures 2.4 cm, so F_3 ≈ 4.8 N at about 117°.

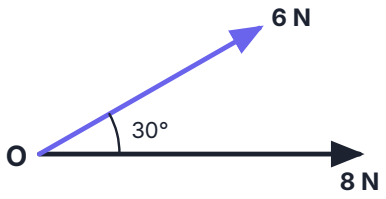
Components: $\Sigma F_y = 6 \sin 70° - 10 = -4.4$ N; $\Sigma F_x = 6 \cos 70° = 2.1$ N

Resultant of F_1 and F_2: $F = \sqrt{2.1^2 + 4.4^2} = 4.9$ N at $\theta = \tan^{-1}(-4.4/2.1) = -64.5°$

F_3 must cancel it: **F_3 = 4.9 N at 115.5°** (from the positive x-axis)

Example 8.1.2

A point object O is acted on by a 6 N force and an 8 N force at 30° to one another. What single third force will balance them?



Solution

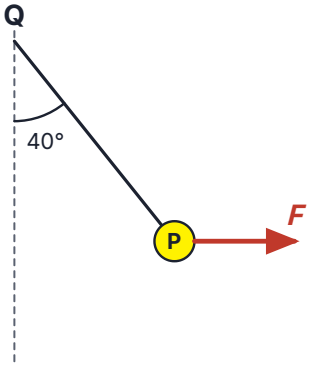
Take the 8 N force along +x: $\Sigma F_x = 8 + 6\cos 30° = 13.20$ N; $\Sigma F_y = 6 \sin 30° = 3.00$ N

Resultant = $\sqrt{13.20^2 + 3.00^2}$ = 13.5 N at 12.8°

The balancing force is equal and opposite: **13.5 N at 192.8°**

Example 8.1.3

A 5 kg body P hangs from a light string fixed at Q. A horizontal force is applied to P, and in equilibrium PQ makes 40° with the vertical. Find the horizontal force and the tension in the string.



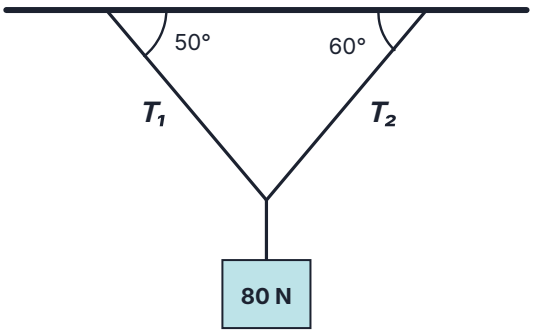
Solution

Resolve vertically: $T\cos 40° - mg = 0 \Rightarrow T = \dfrac{(5)(9.81)}{\cos 40°}$ = **64 N**

Resolve horizontally: $F - T\sin 40° = 0 \Rightarrow F = 64 \sin 40°$ = **41 N**

Example 8.1.4

The object shown weighs 80 N and is in equilibrium. Find T_1 and T_2.



Solution

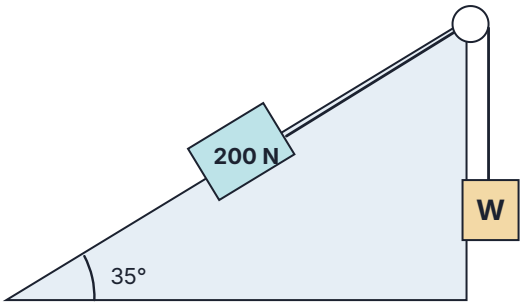
$\Sigma F_x = 0$: $T_2 \cos 60° - T_1 \cos 50° = 0$ (1)

$\Sigma F_y = 0$: $T_2 \sin 60° + T_1 \sin 50° - 80 = 0$ (2)

Solving: $T_1 = 42.5$ N, $T_2 = 54.8$ N

Example 8.1.5

The pulley is frictionless and the friction between the block and the incline is negligible. Find the weight W needed to keep the 200 N block at rest on the 35° incline.



Solution

For the 200 N block, along the incline: $T - 200 \sin 35° = 0 \Rightarrow T = 114.7$ N

For W: $T - W = 0 \Rightarrow$ **$W = 114.7$ N**

Exercises: equilibrium of a particle

1. A wooden block of weight 7.2 N rests on a slope at 20° to the horizontal. Calculate (a) the frictional force on the block, (b) the normal force on the block, (c) the coefficient of static friction.

   Answer

   (a) 2.46 N (b) 6.77 N (c) 0.36

   FULL SOLUTION

   The block is at rest, so the forces along and perpendicular to the slope balance.

   (a) f = W sin 20° = 7.2 × 0.342

   = **2.46 N** (up the slope)

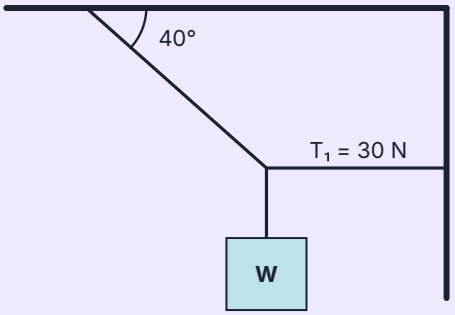
   (b) N = W cos 20° = 7.2 × 0.940

   = **6.77 N**

   (c) Taking the block as on the point of slipping: μ_s = f/N = 2.46 / 6.77 (= tan 20°)

   = **0.36**

2. An object hangs from a knot. One rope pulls horizontally with T_1 = 30 N; the other goes up to the ceiling at 40° to the ceiling. Find the weight of the object.



Answer

W = 30 tan 40° = 25.2 N. (The slide answer, 30 N, would need the angle to be 45°.)

FULL SOLUTION

At the knot: horizontal rope T_1 = 30 N; the other rope T_2 is at 40° to the ceiling (40° above the horizontal).

Horizontal: $T_2 \cos 40° = T_1 = 30 \Rightarrow T_2 = 39.2$ N

Vertical: $T_2 \sin 40° = W$

$W = 30 \tan 40°$

= **25.2 N**

Note: the slide answer of 30 N would need the angle to be 45°.

3. A 40 N weight hangs from a knot held by a rope at 60° to the ceiling and a second rope at 70° to the vertical (going down to the wall). Find the two tensions.

Answer

58 N in the rope at 60° to the ceiling; 31 N in the rope at 70° to the vertical

FULL SOLUTION

Rope 1: at 60° to the ceiling, i.e. 60° above the horizontal. Rope 2: at 70° to the vertical, going down to the wall on the other side.

Horizontal: $T_1 \cos 60° = T_2 \sin 70° \Rightarrow T_1 = 1.879T_2$

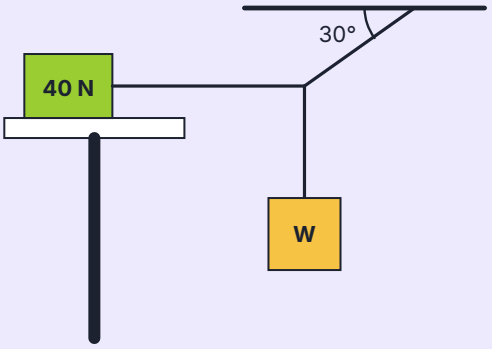
Vertical: $T_1 \sin 60° = 40 + T_2 \cos 70°$

$1.879T_2(0.866) - 0.342T_2 = 40 \Rightarrow 1.286T_2 = 40$

T_2 = **31 N** (rope at 70° to the vertical)

$T_1 = 1.879 \times 31.1$ = **58 N** (rope at 60° to the ceiling)

4. The system shown is in equilibrium. (a) What is the maximum value of W if the friction force on the 40 N block cannot exceed 12.0 N? (b) What is the coefficient of static friction between the block and the table?



Answer

(a) 6.9 N (b) 0.30

FULL SOLUTION

At the knot: horizontal rope to the block (tension T_1), rope to the ceiling at 30° (tension T_2), and W.

Horizontal: $T_1 = T_2 \cos 30°$; vertical: $W = T_2 \sin 30° \Rightarrow W = T_1 \tan 30°$

For the block: the friction balances the horizontal rope, $f = T_1$.

(a) Maximum friction 12.0 N $\Rightarrow T_1 = 12.0$ N

$W_{max} = 12.0 \tan 30°$

= **6.9 N**

(b) At this point the block is about to slip: $\mu_s = f/N = 12.0 / 40$

= **0.30**

5. The same system is just on the point of slipping when $W = 8.0$ N. What is the coefficient of static friction?

Answer

0.35

FULL SOLUTION

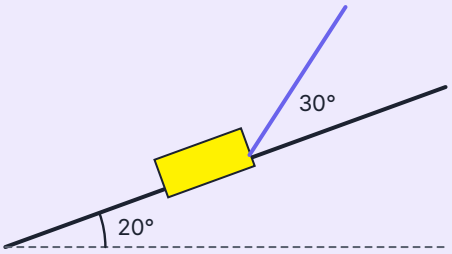
W = 8.0 N, so $T_1 = W / \tan 30° = 8.0 / 0.577 = 13.9$ N

On the point of slipping, friction is at its maximum: $f = \mu_s N = T_1$

$\mu_s = 13.9 / 40$

= **0.35**

6. A 2 kg body rests on a smooth slope at 20° to the horizontal, held by a string at 30° to the slope. Find the tension in the string.



Answer

7.74 N

FULL SOLUTION

Smooth slope: no friction. Resolve along the slope (the normal force has no component there).

$T \cos 30° = mg \sin 20°$

$T = 2(9.81)(0.342) / 0.866$

$= \mathbf{7.74\ N}$

Example 8.1.6

Dynamic equilibrium. A hanging 8.0 N block pulls a 20 N block along a table at constant velocity. Find μ_k between the 20 N block and the table. (Frictionless pulley.)

Solution

Constant velocity, so each block is in equilibrium. Hanging block: $T = 8.0$ N

20 N block: $T = f_k = \mu_k R$ and $R = 20$ N

$$\mu_k = \frac{8.0}{20} = \mathbf{0.40}$$

Practice question · equilibrium on an incline

A box of mass 10.0 kg rests in equilibrium on a smooth plane inclined at 35° to the horizontal, held by a string parallel to the incline. Calculate the tension in the string. ($g = 9.81$ m s^{-2})

Answer

Along the incline: $T = mg \sin 35° = (10.0)(9.81)(0.574) = \mathbf{56.2\ N}$

8.1 Equilibrium of a rigid body

Rigid body

is a body with a **definite and unchanging shape and size**. It can have translational motion and rotational motion.

Torque (moment of a force)

Torque, τ

is the **vector product of r and F**, where **r** is the vector from the axis of rotation to the point where the force acts. It measures the tendency of a force to rotate a body about the axis.

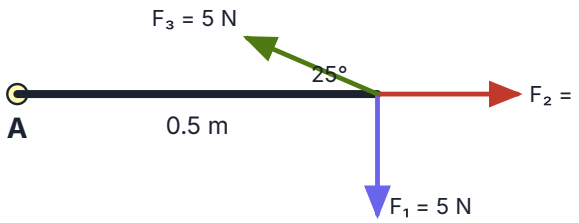
$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} \quad \tau = rF\sin\theta = Fd$$

θ = angle between **r** and **F**; $d = r\sin\theta$ = lever arm (moment arm); unit N m

- The **lever arm** d is the perpendicular distance from the axis to the line of action of the force.
- Torque is a vector; its direction is given by the right-hand rule (the thumb), along the axis.
- Sign convention: **clockwise negative**, **anticlockwise positive**.

Example 8.2.1

Calculate the torque about axis A due to each force shown: F_1 = 5 N downward at 0.5 m from A, F_2 = 5 N along the rod, and F_3 = 5 N at 25° to the rod.



Solution

$\tau = rF\sin\theta$

$\tau_{1A} = -(0.5)(5\sin 90°) = \mathbf{-2.5\ N\,m}$

$\tau_{2A} = (0.5)(5\sin 0°) = \mathbf{0}$ (the line of action passes through A)

$\tau_{3A} = +(0.5)(5\sin 25°) = \mathbf{+1.1\ N\,m}$

Example 8.2.2

Calculate the torque about A due to each force: 10 N downward, 20 N along the rod and 25 N at 65° to the rod, all acting 0.8 m from A.

Solution

$\tau_{1A} = -(0.8)(10\sin 90°) = \mathbf{-8.0\ N\,m}$

$\tau_{2A} = (0.8)(20\sin 0°) = \mathbf{0}$

$\tau_{3A} = +(0.8)(25\sin 65°) = \mathbf{+18.1\ N\,m}$

Example 8.2.3

The torque about X is 25 N m. Find the angle θ when (a) an 8 N force acts 4 m from X, (b) a 60 N force acts 0.5 m from X.

Solution

(a) $25 = (4)(8 \sin\theta) \Rightarrow \sin\theta = 0.781 \Rightarrow \mathbf{\theta = 51.4°}$

(b) $25 = (0.5)(60 \sin\theta) \Rightarrow \sin\theta = 0.833 \Rightarrow \theta = 56.4°$ or **124°** (the angle in the figure is obtuse)

Conditions for the equilibrium of a rigid body (coplanar forces)

$\Sigma \mathbf{F} = 0 \quad (\Sigma F_x = 0, \Sigma F_y = 0)$

1st condition (force condition): no translational motion

$\Sigma\tau = 0$ about any axis

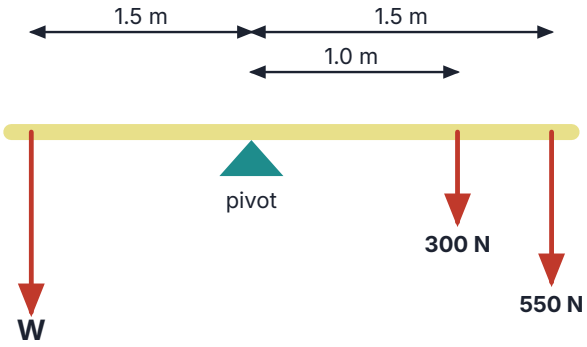
2nd condition (torque condition): no rotational motion

Teacher's note

Choose the axis for $\Sigma\tau = 0$ at the point where an unknown force acts (a hinge, a pivot, the foot of a ladder): that force then has zero torque and disappears from the equation.

Example 8.2.4

The see-saw shown is balanced. Calculate the weight W.



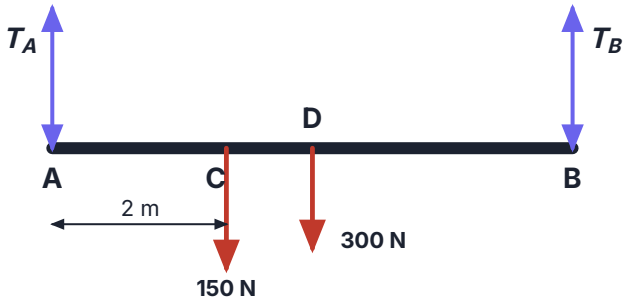
Solution

$\Sigma\tau = 0$ about the pivot: $(1.5)(W \sin 90°) - (1.0)(300 \sin 90°) - (1.5)(550 \sin 90°) = 0$

$\boldsymbol{W = 750\ \text{N}}$

Example 8.2.5

A uniform plank AB, 6 m long and of weight 300 N, is held horizontal by two vertical ropes at A and B. A 150 N weight rests on the plank at C, where AC = 2 m. Find the tension in each rope.



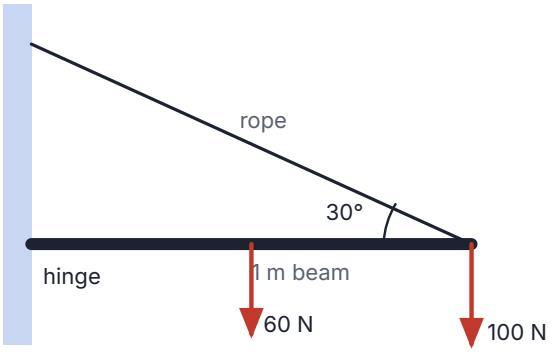
Solution

Take A as the axis: $\Sigma\tau_A = 0$: $(0)T_A - (2)(150) - (3)(300) + (6)T_B = 0 \Rightarrow \boldsymbol{T_B = 200\ \text{N}}$

$\Sigma F_y = 0$: $T_A + T_B - 150 - 300 = 0 \Rightarrow \boldsymbol{T_A = 250\ \text{N}}$

Example 8.2.6

A uniform 1 m beam of weight 60 N is hinged to a wall and held horizontal by a rope at 30° to the beam, attached at its end. A 100 N load hangs from the end. Calculate (a) the tension in the rope, (b) the force of the hinge on the beam.



Solution

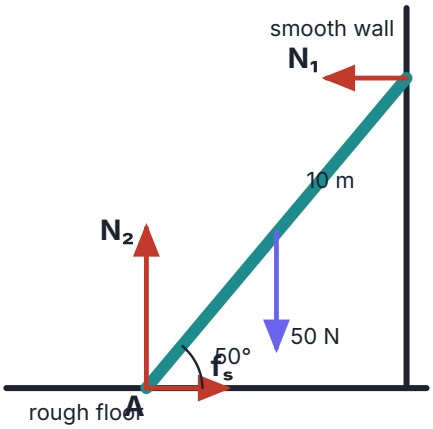
(a) $\Sigma\tau = 0$ about the hinge: $-(0.5)(60) - (1)(100) + (1)(T\sin 30°) = 0 \Rightarrow$ **$T = 260$ N**

(b) $\Sigma F_x = 0$: $R_x = T\cos 30° =$ **225 N**; $\Sigma F_y = 0$: $R_y = 160 - T\sin 30° =$ **30 N**

Hinge force = $\sqrt{225^2 + 30^2}$ = **227 N** at $\tan^{-1}(30/225) = 7.6°$ above the horizontal

Example 8.2.7

A 10 m ladder of weight 50 N leans against a smooth wall at 50° to the rough floor. (a) How large a friction force must act at the foot so that it does not slip? (b) What coefficient of static friction is needed?



Solution

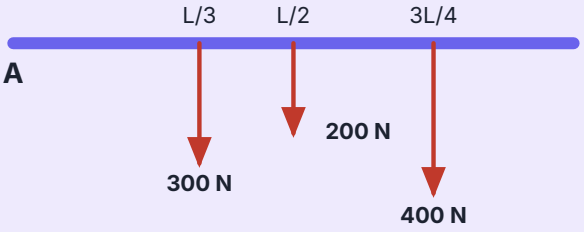
$\Sigma F_x = 0$: $f_s = N_1$ (1) ; $\Sigma F_y = 0$: $N_2 = 50$ N (2)

$\Sigma\tau = 0$ about the foot A: $-5(50 \sin 40°) + 10(N_1 \sin 50°) = 0 \Rightarrow N_1 = 21$ N

(a) f_s **= 21 N** (b) $f_s = \mu_s N_2 \Rightarrow \mu_s = \dfrac{21}{50}$ = **0.42**

Exercises: equilibrium of a rigid body

1. A uniform board of weight 0.20 kN and length L has 300 N hanging at $L/3$ from end A and 400 N at $3L/4$ from A. What single additional force will keep the board in equilibrium?



Answer

0.90 kN upward, at 0.56L from A

FULL SOLUTION

Weights: 300 N at L/3, board 200 N at L/2, 400 N at 3L/4 (all measured from A).

Translational equilibrium: F = 300 + 200 + 400

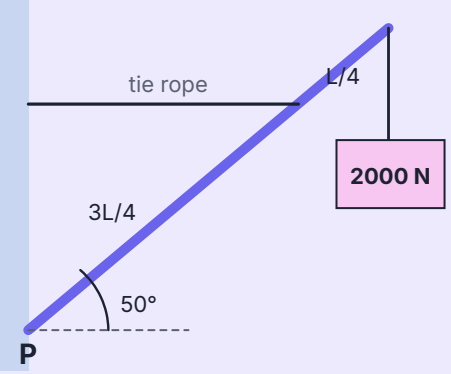
= **900 N = 0.90 kN upward**

Rotational equilibrium (moments about A): F x = 300(L/3) + 200(L/2) + 400(3L/4) = 100L + 100L + 300L = 500L

x = 500L / 900

= **0.56L from A**

2. A uniform 0.40 kN beam is hinged at P and held at 50° to the horizontal by a horizontal tie rope attached $3L/4$ along it; a 2000 N load hangs from its end. Calculate the tension in the rope and the force of the hinge on the beam.



Answer

T = 2460 N; hinge force 3.4 kN at 44° to the horizontal

FULL SOLUTION

Beam weight 400 N at L/2; load 2000 N at the end L; rope horizontal at 3L/4; beam at 50° to the horizontal.

Moments about P (perpendicular distances):

T(3L/4) sin 50° = 400(L/2) cos 50° + 2000(L) cos 50°

T = (200 + 2000) cos 50° / (0.75 sin 50°) = 1414 / 0.5745

= **2460 N**

Hinge: horizontal component H = T = 2460 N; vertical component V = 400 + 2000 = 2400 N

R = $\sqrt{(2460^2 + 2400^2)}$ = 3.44×10^3 N

= **3.4 kN**

tan φ = V/H = 2400 / 2460 ⇒ φ = **44° to the horizontal**

Practice question · centre of gravity and toppling

A rectangular block 4 cm wide and 8 cm long stands on a plane inclined at angle θ. Which statement is true?

A. The block topples if θ = 25°.

B. The block does not topple because it has a higher centre of gravity.

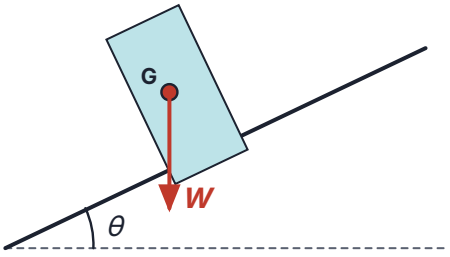
C. The block does not topple while the line of action of its weight passes inside its base (through or before the pivot).

D. The block topples if its length lies along the plane at the same angle.

Answer

Answer: C

The block tips over the lower edge when the vertical line through its centre of gravity passes outside the base, i.e. when $\tan\theta > \frac{4}{8}$, $\theta > 26.6°$. While the weight acts inside the base, its moment about the pivot edge restores the block. At θ = 25° it does not topple.



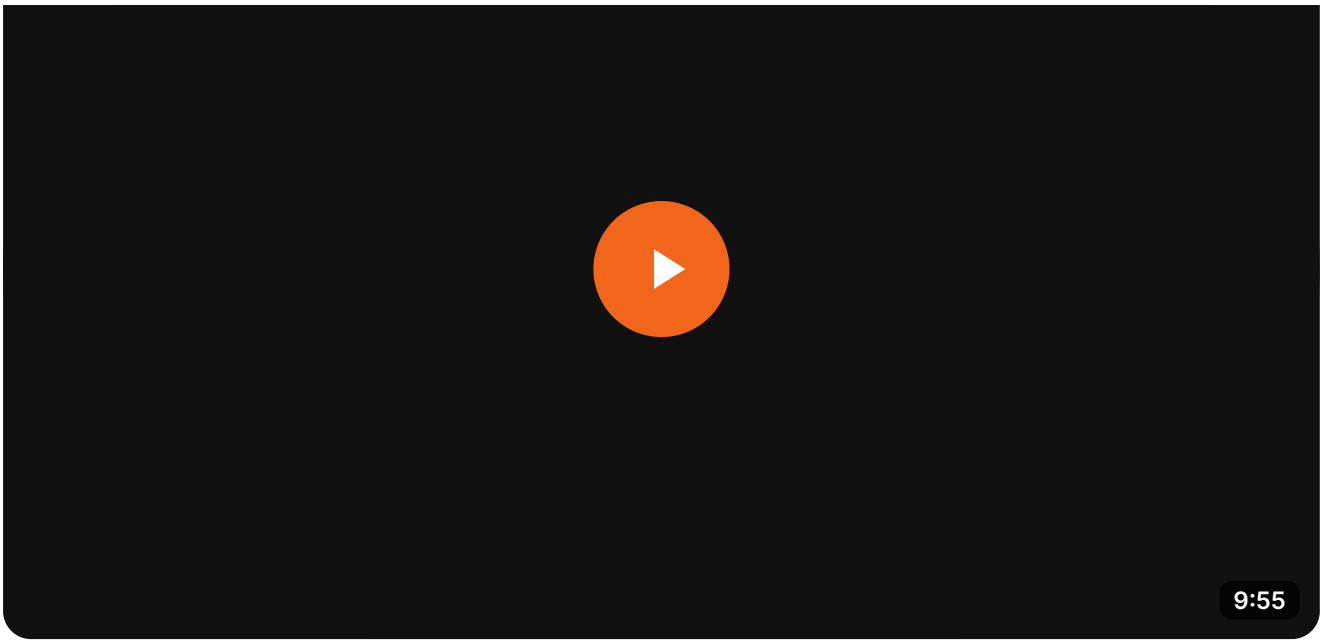
Real-World Physics: why the Burj Khalifa doesn't fall over

REAL-WORLD PHYSICS Dubai · Richard Hammond's BIG

Standing 828 m tall on sand and weak rock

The Burj Khalifa in Dubai is 828 m tall, the tallest building on Earth. It stands on sand and weak rock, and must resist strong desert winds. In *Richard Hammond's BIG*, Richard investigates how it stays standing.

Its weight rests on a thick concrete raft supported by about 190 deep concrete piles, which hold the tower up mainly by **friction** with the surrounding ground. Its Y-shaped plan, with three wings bracing a central core, gives a wide base, and the tower steps inwards ("setbacks") as it rises.



The Physics Behind The Burj Khalifa And Why It Doesn't Fall Over · Quest TV

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While you watch: identify the forces acting on the tower (weight, upward forces from the piles, wind) and how the design keeps it in equilibrium.

The physics

- For a rigid body in **equilibrium**: the resultant force is zero *and* the resultant torque about any point is zero.
- The upward friction and normal forces from the piles balance the weight of the tower.
- A wide base and a low **centre of gravity** make the tower stable: the wind would have to tilt it a long way before the line of action of its weight moved outside the base.

Quick check 1

What are the conditions for the Burj Khalifa to be in static equilibrium?

A The resultant force on it is zero only.

B The resultant torque on it is zero only.

C The resultant force on it is zero and the resultant torque about any point is zero.

D Its weight is equal to the wind force.

Quick check 2

Assume the weight of the tower is 4.8×10^9 N and it is shared equally by 192 piles. What is the upward force provided by each pile?

A 2.5×10^5 N

B 2.5×10^6 N

C 2.5×10^7 N

D 9.2×10^{11} N

Quick check 3

Why does a wide base and a low centre of gravity make a tall building more stable against wind?

A They make the weight of the building smaller.

B The line of action of the weight stays within the base for a larger tilt, so the weight gives a restoring torque.

C They reduce the wind force to zero.

D They move the centre of gravity outside the base.

Quick check 4

(Link to Chapter 9) A pile of cross-sectional area 1.77 m^2 carries a force of 2.5×10^7 N. What is the compressive stress in the pile?

A 4.4×10^7 Pa

B 1.4×10^7 Pa

C 7.1×10^{-8} Pa

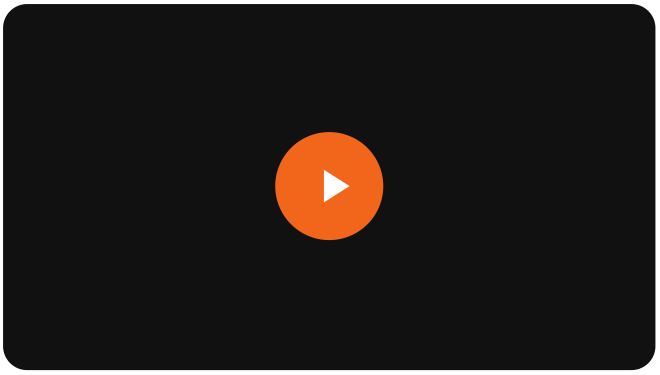
D 1.4×10^5 Pa

Source: *Richard Hammond's BIG*, Episode 7 "Tallest Building On Earth" (Discovery, 2020). Values in the calculations are rounded estimates for practice. Preview the clip before class. youtube.com/watch?

v=N5i3UsiSoYY

Physics in Real Life

Real-world videos for this chapter. Tap a picture to play it here.



ENGINEERING EXPLAINER

How Cranes Lift Heavy Loads Without Tipping Over

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How a tower crane stays upright

The counterweight balances the moment of the load about the tower: for equilibrium the sum of clockwise moments equals the sum of anticlockwise moments, and the centre of gravity must stay over the base.